Adaptive time-stepping for variational quantum imaginary-time evolution: report on the work completed

Variational quantum imaginary-time evolution (QITE) integrates the linear system $M\dot{\theta} = Y$ given by McLachlan's principle. Both the step size $\Delta\tau$ and the conditioning of the metric M decide whether a run is accurate and how much it costs. This report collects, phase by phase, what the project has done through Phase V and what it found. All results are for H_2 in STO-3G on two qubits. The work falls into five phases. *(I)* Replicated Saxena, Shalabi and Kyaw's noise-scaling analysis. Theorem 1 and the noisy-flow scaling reproduce to 10^{-10}. The quoted condition numbers do not reproduce (863.26 against 66.72), and the 9-parameter metric is exactly rank 6 of 9. It was found that Theorem 2 overstates the hardware error by exactly a factor of two. *(II)* An Adam-like step controller was built and audited. Read at the circuit budget where the simpler ratchet controller finishes, it wins in every case tested; at 732 circuits the two differ by only 3×10^{-5}. Two headline claims were withdrawn. One positive result was confirmed: a zero-cost local-error estimator tracks step doubling with log-Pearson correlation 0.9902, at the predicted ratio of 2. *(III)* A physics-indicator controller (Route 3) was implemented and audited. At a matched budget of 732 circuits it lands 2.2× closer to the variational floor than fixed-step Euler. A debiasing "surprise" was traced to a wrong bias factor and retracted, and so was a claimed noise floor. *(IV)* Replication of Assi, Vogl, Kumari and LeBlanc's variational product formulas. Their figures reproduce to $\lesssim$1.2%, but four internal inconsistencies were found in the paper, and the unreported metric condition number reaches 10^{16} in a boundary layer. *(V)* Two pre-registered experiments followed. They show that the Route-3 stiffness is a chart singularity of the parametrization, not a numerical error. They also show that the end-point error does not track a requested tolerance, for reasons identified. Every number quoted is pinned by a verification script that has no tolerance parameter, except a few single-run checks labelled as such (Sec. IX).

I. INTRODUCTION

Imaginary-time evolution $|\psi(\tau)\rangle \propto e^{-H\tau}|\psi_0\rangle$ drives any state with nonzero ground-state overlap to the ground state of H [4, 5]. The variational form of QITE restricts the state to a parametrized circuit $|\psi(\theta)\rangle$. McLachlan's principle then gives an ordinary differential equation for the parameters [1, 3, 6],

$$M(\theta)\,\dot{\theta} = Y(\theta), \tag{1}$$

with M the (real part of the) quantum geometric tensor and Y an energy-gradient vector. On hardware each evaluation of $\dot{\theta} = f(\theta) = M^{+}Y$ costs $O(N^2)$ circuits for N parameters. The step size therefore sets the circuit bill directly. The quality of the linear solve depends on $\mathrm{cond}(M)$ and on the cutoff used in the pseudo-inverse.

The project starts from *Adaptive Time-Stepping for Variational QITE: An RKF-Type Controller with a Hardware-Noise Floor* [7], hereafter the memo. The memo treats Eq. (1) as an ODE integrated by a Runge–Kutta–Fehlberg-type controller, and it adds a noise floor derived from the Saxena–Shalabi–Kyaw (QDP) bounds [1]. It proposes five attack routes. This report documents what has been built, measured, audited and corrected on top of it. It is organized by phase (Table I).

Main established results.

1. The QDP noise-scaling results (Theorem 1 and the noisy-flow scaling) reproduce to about 10^{-10}. On hardware, where the state inside $\{H, \rho\}$ is also noisy, the parameter-velocity error is *exactly half* of what Theorem 2 states (Sec. III).

2. The QDP ansätze are far worse conditioned than reported. The 5-parameter metric has $\mathrm{cond}_F = 863.26$, not 66.72. The 9-parameter metric is exactly rank 6 of 9, which is structural because a two-qubit pure state has $2d - 2 = 6$ real degrees of freedom.

3. A zero-cost local-truncation-error estimator is already available in any energy-gated controller. It agrees with step doubling at the predicted ratio of 2 (Sec. IV).

4. Adding momentum (Adam-style EMAs) to the step-size law does not beat a simple ratchet at matched circuit budget (Sec. IV).

5. At a matched budget of 732 circuits, a cap based on $\mathrm{cond}(M)$ plus a variational energy gate lands 2.2× closer to the variational floor than fixed-step Euler (Sec. V).

6. The large-$\mathrm{cond}(M)$ events in the 9-parameter runs are a chart singularity. The parameter speed diverges while the state speed stays bounded. They are not gauge leakage and not finite-difference noise (Sec. VII).

7. With a proportional–integral (PI) law and an explicit tolerance, the controller does regulate its local-error estimate. The error at the final time still does not scale with the tolerance, because the flow is contractive and the first step is fixed (Sec. VII).

TABLE I. The project by phase. "Pinned" counts the numerical values checked by the phase's verification script (Appendix A).

Phase	Work	Outcome
I	Replication of Saxena–Shalabi–Kyaw [1] (T1–T5)	T1, T4 reproduced; T2, T5 partially; T3 not reproduced. Factor-of-two correction to Theorem 2.
II	Adam-like step controller (E1–E3); audit G1–G7	Ratchet wins at matched completion budget; E2 withdrawn, E3 re-attributed; zero-cost error estimator validated; 147 values pinned, 0 drift.
III	Route-3 controller (Targets A–D); audit F1–F12, modules M1–M6	2.2× closer to floor at 732 circuits; debias factor corrected; tight `rcond` beats loose by 2.4×; noise-floor claim retracted.
IV	Replication of Assi–Vogl–Kumari–LeBlanc [2] (T1–T8)	Figures reproduce; four internal inconsistencies found; cond(g) boundary layer (T6) pinned.
V	P1 (singularity classification); P2 (tolerance proportionality)	Chart singularity, no gauge leakage (51 pinned); no tolerance fit possible, post-hoc first-step ablation (93 pinned).

II. FRAMEWORK AND METHODS

A. Variational QITE and the linear solve

For a density matrix $\rho(\theta)$ the metric and gradient are [1]

$$M_{kq} = \mathrm{Tr}[\partial_k\rho\,\partial_q\rho]\,, \tag{2}$$

$$Y_k = -\mathrm{Tr}[\partial_k\rho\,(\{H,\rho\} - 2\langle H\rangle\rho)]\,. \tag{3}$$

For a pure state $\partial_k\rho = |\partial_k\psi\rangle\langle\psi| + |\psi\rangle\langle\partial_k\psi|$, and normalization makes $\langle\partial_k\psi|\psi\rangle$ purely imaginary. Equation (3) then collapses to

$$Y_k = -\partial_k E, \qquad E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle, \tag{4}$$

which was verified numerically to $\|Y+\nabla E\|/\|Y\| = 2.3\times 10^{-10}$. For a product of Pauli rotations with generators $\tilde{G}_k$ (dressed by the gates to their left),

$$\begin{aligned} M_{jk} &= 2\left(\mathrm{Re}\langle\tilde{G}_j\tilde{G}_k\rangle - \langle\tilde{G}_j\rangle\langle\tilde{G}_k\rangle\right), \\ Y_k &= -i\langle[\tilde{G}_k, H]\rangle, \end{aligned} \tag{5}$$

which was checked against the finite-difference M and Y to 2.594×10^{-10} and 3.955×10^{-10} (Sec. VII).

Equation (1) is integrated with explicit Euler, $\theta_{n+1} = \theta_n + \Delta\tau_n\, M^{+}Y$. The pseudo-inverse cutoff `rcond` (relative to the largest singular value) is always an explicit, logged parameter, and the reported cond(M) is always stated as either the regularized Frobenius value cond_F or the unregularized spectral value. Section VII shows that this distinction is significant.

B. Noise model and the QDP bounds

Following Ref. [1], every gate is followed by the global depolarizing channel $\mathcal{E}(\rho) = (1-p)\rho + p\,\mathbb{I}/d$, with $d = 4$. After N gates, and writing $q \equiv (1-p)^N$,

$$\rho^{\mathrm{err}} = q\,\rho + (1-q)\,\mathbb{I}/d. \tag{6}$$

Theorem 2 of Ref. [1] states that the relative velocity error is $\epsilon/\|\dot\theta\| = (1-q)/q$, independent of the Hamiltonian. Theorem 1 bounds the same error in terms of $\|M\|$, $\|M^{-1}\|$ and $\|Y\|$. The memo inverts the Theorem-2 bias as $\tilde f = q^{-1}f$ (memo Eq. 7) and combines truncation and residual error into

$$e_n^2 \approx C_{\mathrm{tr}}^2\Delta\tau^{2(p+1)} + \Delta\tau^2\kappa_b^2\varepsilon_{\mathrm{res}}^2\|\dot\theta\|^2, \tag{7}$$

where p is the order of the integrator (memo Eq. 8). Below the resulting $\Delta\tau_{\mathrm{floor}}$, shrinking the step does not help (memo Eq. 9). The memo's Route 3 replaces the RKF error estimate with physics indicators: the McLachlan residual, energy monotonicity $dE/d\tau = -2\,\mathrm{Var}_\rho(H)$ (memo Eq. 13), and cond(M) as a stiffness proxy. Its Route 4 proposes the PI law

$$\Delta\tau_{n+1} = \Delta\tau_n\left(\frac{\mathrm{tol}}{e_n}\right)^{k_I}\left(\frac{e_{n-1}}{e_n}\right)^{k_P}, \tag{8}$$

with $k_I \approx 0.3/(p{+}1)$ and $k_P \approx 0.4/(p{+}1)$ (memo Eq. 17).

C. Baseline step law (Route 3)

All controllers in this report are compared with a common baseline, which is called Route 3 throughout this document. It is the implementation of the memo's third route. The memo gives no numbers and no rule for combining the indicators, so both the rule and every constant in Table II are heuristic choices. Each iteration does the following.

1. **Stiffness cap (proactive).** Measure M and Y and solve. Then set $\Delta\tau \leftarrow \max\big(\Delta\tau_{\min}, \min(\Delta\tau,\, c_{\mathrm{cap}}/\mathrm{cond}(M),\, \Delta\tau_{\max})\big)$. A singular M (cond $= \infty$) sends $\Delta\tau$ to the floor, not to zero.
2. **Energy gate (reactive).** Take the trial step and compare the energy change ΔE_a with the variational first-order prediction $\Delta E_p = -(Y\cdot\dot\theta)\Delta\tau$.

If E rises by more than a monotonicity tolerance, or if $|\Delta E_a - \Delta E_p|$ exceeds $\max(\mathrm{ABS}, \mathrm{REL}\,|\Delta E_p|)$, shrink $\Delta\tau$ by 1/2 and retry, up to 20 times. If the retries run out, the last trial is committed.

3. **Ratchet.** After an accepted step, multiply $\Delta\tau$ by 1.25 unless the relative McLachlan residual exceeds 5%, in which case growth is frozen. The residual never triggers a retry.

The cap coefficient was first set to 50. At that value the cap dominated the whole run, so it was raised to 2000. The gate originally tested against the exact-ITE identity $-2\,\mathrm{Var}(H)\Delta\tau$. The audit (F5) replaced this with the variational prediction, because the ansatz deficit does not vanish at convergence. The baseline described here is the post-audit version.

D. Benchmark

All runs use H_2 in STO-3G under the Bravyi–Kitaev mapping, reduced to two qubits, with the 5-parameter and 9-parameter ansätze of Ref. [1]. Derivatives $\partial_k\rho$ are central finite differences with $h = 10^{-6}$. They are validated against the closed-form diagonal $M_{kk} = \frac{1}{2}(1 - \mathrm{Tr}[P_k\rho P_k\rho])$ (Eq. B9 of Ref. [1]), with a worst deviation of 8.9×10^{-11}. Simulations use PennyLane 0.45.1 with fixed seeds, and each target has its own seed. $E_{\mathrm{floor}} = -1.124600$ is the variational minimum of the 5-parameter ansatz, obtained by direct minimization. It lies 2.100×10^{-2} above the exact ground-state energy. Convergence against E_{floor} is reported because no controller can get past it.

E. Evidence protocol

Every figure, table entry or claimed number has its own function in a script. Verdicts use three words: *reproduced*, *partially reproduced* and *not reproduced*. Every quoted value is pinned in a verification script, and a `-check` mode re-runs the computation and compares against the pinned values with no adjustable tolerance. From Phase V on, predictions were registered in writing before the run, and their outcomes are scored as held or failed. Controllers are compared in a common currency of *circuits*. One evaluation of f with N parameters costs $2N+1$ circuits, and each energy-gate trial, including the first, costs one more. Comparisons "at matched budget" read every run at the same circuit count.

TABLE II. Constants of the baseline step law. None comes from the memo; all were chosen for this work.

Constant	Value	Role
$\Delta\tau_0$	0.05	initial step
$\Delta\tau_{\min}$, $\Delta\tau_{\max}$	10^{-4}, 0.5	clip range
GROW, SHRINK	1.25, 0.5	ratchet, retry
c_{cap}	2000	stiffness cap $c_{\mathrm{cap}}/\mathrm{cond}(M)$
`rcond` (default)	10^{-6}	pseudo-inverse cutoff
monotonicity tol.	10^{-9}	energy may not rise
ABS, REL	10^{-3}, 0.5	energy-gate band
residual tol.	0.05	freezes growth
max. retries	20	per step

TABLE III. Phase I verdicts for Ref. [1].

Target	Verdict
T1: Theorem 1 bound, Eq. (17)	*reproduced*
T2: $p_{\max}$ at $\delta = 0.04$, $N = 5$–14	*partially reproduced*
T3: cond(M), 5- and 9-param. ansätze	*not reproduced*
T4: noisy flow vs Eq. (19)	*reproduced* ($\sim 10^{-10}$)
T5: Fig. 3(b) Theorem-1 curves	*partially reproduced*

TABLE IV. T2: $p_{\max}$ at $\delta = 0.04$.

N	paper	this work	deviation
5	0.032	0.03311	+3.48%
6	0.036	0.03653	+1.48%
10	0.052	0.05349	+2.86%
12	0.068	0.06964	+2.41%
14	0.129	0.13083	+1.42%

III. PHASE I: REPLICATION OF THE QDP NOISE ANALYSIS

Question. Does the noise-scaling analysis on which the memo builds its floor reproduce, and does it apply as stated to hardware?

Summary. Table III lists the targets. The closed-form results reproduce, while the numbers that depend on unstated details do not.

T2: maximal noise rate. All five values of $p_{\max}$ come out between 1.4% and 3.5% *above* the paper's (Table IV). The offset is one-sided. Back-solving $\|M\|$ from each paper value gives 1.1018, 1.0456, 1.0947, 1.0631 and 1.0075, which are mutually inconsistent. A common rescaling of $\|M\|$ therefore cannot explain the gap. The pattern is consistent with the paper quoting the last point of a discrete p grid that still satisfies the bound.

T3: conditioning of the ansätze. For the 5-parameter ansatz the replication gives $\mathrm{cond}_F = 863.2577$ against the paper's 66.7239 (+1194%). The corresponding values are $\|M\|_F = 0.8793$ (paper 0.9977) and spectral condi-

TABLE V. T5: Fig. 3(b) read-offs against three readings of Eq. (17).

	N=5	6	10	12	14
Fig. 3(b) read-off	75	105	300	430	590
Eq. (17) as printed	103.4	146.6	396.6	575.3	824.3
Eq. (17) per (A37)	119.0	168.8	457.0	665.9	969.5
$\|Y\|$ term dropped	77.2	109.7	295.7	423.9	581.7

tion number 624.9. Because a condition number is scale-free, no normalization convention can close this gap. A 60-row scan over the choices the paper leaves open (initial state, controlled or Ising two-qubit gate, wire order, and three metric conventions) gives values clustered at 863.26, 879.59 and 911.08. None of them lands within 5% of the paper. The 9-parameter ansatz is worse. Its singular values are

$$[1.267,\ 0.706,\ 0.633,\ 0.383,\ 0.181,\ 0.0432,\ 8.8\times10^{-17},\ 2.9\times10^{-17},\ 3.2\times10^{-18}],$$

so M has rank 6 and $\mathrm{cond}(M) = \infty$, where the paper reports 20.22. The rank deficiency is structural. The pure states of two qubits form a manifold of real dimension $2d - 2 = 6$, so any 9-parameter ansatz carries at least three gauge directions. A finite value such as the paper's is therefore a property of the pseudo-inverse cutoff, not of the ansatz. The regularized value is 39.41 for every `rcond` from 10^{-15} to 10^{-2}, then 10.69 at 0.1 and 5.66 at 0.3. None matches 20.22. The same ansatz gives $\|M\|_F = 1.6387$ (paper 1.689) and $\|Y\| = 0.1927$ (paper 0.121).

T4: the noisy flow. Simulating the circuit with a depolarizing channel after every gate reproduces Eq. (19) of Ref. [1] to about 10^{-10}. The residuals are $M^{\mathrm{err}} \le 4\times10^{-10}$ and $Y^{\mathrm{err}} \le 10^{-9}$, with a largest $\dot\theta$ deviation of 2.8×10^{-7} (Fig. 1, circles).

T5: Theorem-1 curves. Fig. 3(b) of Ref. [1] can be read in at least three ways (Table V). The reading that drops the $\|Y\|$ term is closest to the figure. None matches it at every N. The denominator of Eq. (A42) is also positive only for $p \le 4.28\times10^{-4}$ ($N = 5$) down to $p \le 6.68\times10^{-5}$ ($N = 14$), which is below most of the plotted range.

New result: Theorem 2 is a factor of two pessimistic on hardware. The derivation in Appendix B of Ref. [1] applies noise to the derivative $\partial_k\rho$ but keeps the state in $\{H,\rho\}$ ideal. On hardware both come from the same noisy device. Using Eq. (6) for both, with $\mathrm{Tr}\,\partial_k\rho = 0$ and $\mathrm{Tr}[\partial_k\rho\,\rho] = 0$ (Appendix B 1), gives

$$M^{\mathrm{err}} = q^2 M, \qquad Y^{\mathrm{err}} = \Big[q^2 + \tfrac{2}{d}q(1-q)\Big]Y, \tag{9}$$

$$\dot\theta^{\mathrm{err}} = \dot\theta\left[1 + \frac{1-q}{2q}\right] \quad (d=4), \tag{10}$$

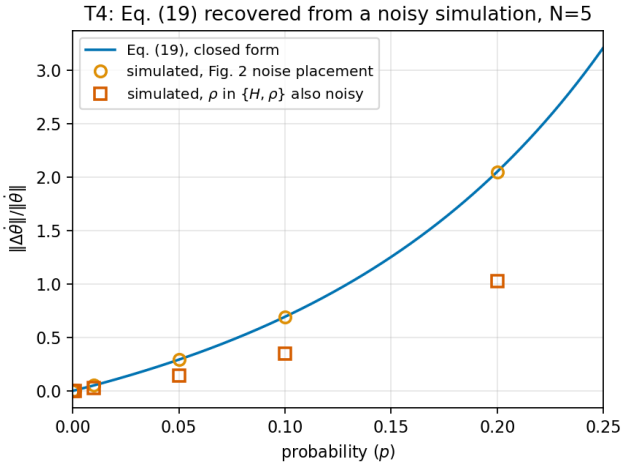


FIG. 1. T4 and the factor of two ($N = 5$). The line is the closed form of Eq. (19) in Ref. [1]. Circles are the simulated relative velocity error with noise placed as in the paper's Fig. 2, where only $\partial_k\rho$ is noisy. Squares have the state inside $\{H,\rho\}$ noisy as well, as it is on hardware. The squares sit at exactly half the closed form.

so the relative error is $(1-q)/(2q)$, exactly half of Theorem 2's $(1-q)/q$. Numerically, the ratio of the two errors equals 0.5 to within 3.8×10^{-5} across $p \in [10^{-4}, 0.2]$ (Fig. 1, squares). The consequence for the memo is concrete: its debiasing factor q^{-1} (memo Eq. 7) over-corrects on hardware. The correct factor is $\big[q^2 + q(1-q)/2\big]/q^2$. Phase III first missed this and then corrected it.

IV. PHASE II: ADAM-LIKE STEP CONTROLLER AND ITS AUDIT

Question. Can the step-size law of the baseline (Sec. II C) be improved by adding the memory of the Adam optimizer, that is, by smoothing the error signal before acting on it?

A. Design

The controller keeps Route 3's cap, gate and thresholds and replaces only the ratchet. It uses the energy-gate discrepancy, normalized by the gate band, as an error signal:

$$\hat e_n = \frac{|\Delta E_a - \Delta E_p|}{\max(\mathrm{ABS},\ \mathrm{REL}\,|\Delta E_p|)}. \tag{11}$$

Bias-corrected exponential moving averages $\hat m_e$ and $\hat v_e$ of $\hat e$ and $\hat e^2$ ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$) define a confidence ratio $s_e = \hat m_e/(\sqrt{\hat v_e} + \varepsilon)$. The growth factor is

$$\mathrm{growth} = \max(\hat m_e, \varepsilon)^{-s_e k_I}, \qquad k_I = 0.15, \tag{12}$$

clipped to $[0.5, 1.25]$. A parallel EMA of the residual only sets a ceiling on growth. The Adam machinery acts on the scalar $\Delta\tau$ alone and never rescales individual parameters. A per-parameter rescaling would change the ODE being integrated, which would make it a different method, not a step controller for the same one.

Consistency check. With $\beta_1 = \beta_2 = 0$ and $k_I = 10$, the controller reproduces Route 3 exactly (172 steps, final $E = -1.122993$). A memoryless variant gives 172 steps and -1.122928.

B. Experiments as first reported

Three experiments were run. E1 compares convergence on the noiseless 5-parameter ansatz (Fig. 2). E2 compares behaviour under depolarizing noise. E3 compares behaviour on the rank-deficient 9-parameter ansatz (Fig. 3). The first report credited Adam with a small E1 advantage at 732 circuits (-1.121027 against -1.120992), with robustness to noise in E2, and with a better cap-off result in E3.

C. Audit (G1–G7) and what it changed

A full audit re-ran every claim against live code. It produced seven findings, G1–G7. The consequential ones follow.

G2: the β_2 horizon decides every verdict. With $\beta_2 = 0.999$ the second moment has an effective memory of about 1000 steps, longer than any run. $\sqrt{\hat{v}_e}$ is then set by a much longer memory than $\hat{m}_e$, and s_e collapses toward zero. The "frozen" fraction in Table VI counts the steps where this happens. The table repeats E1 on two seeds at matched circuit budget, for the stock setting and for three settings with equal horizons. *Read at Route 3's completion budget (2069 and 2816 circuits), Route 3 wins in every case.* The stock Adam arm on the second seed does not reach $\tau_{\max}$ within 500 steps. With equal β's, Adam ends slightly below Route 3 in final energy, but only after spending more circuits.

The horizon sweep on seed D (Table VII) makes the mechanism visible. Every equal-horizon arm reaches $\tau_{\max}$ in about Route 3's number of steps. Reversing the horizons ($\beta_1 = 0.999$, $\beta_2 = 0.9$) drives the mean s_e to 258. This also shows that the bound $s_e \leq 1$ stated in the first report is false. Even in the stock E1 run s_e spans $[5.097 \times 10^{-5}, 1.015]$.

G4: the law cannot shrink the step. In E1, $\hat{e} > 1$ on 0 of 190 steps (maximum 0.945), and growth < 1 on 0 of 190. This is not a tuning problem but an identity. The error in Eq. (11) is normalized by the gate's own reject band, so $\hat{e} > 1$ *is* the reject condition. The retry loop shrinks $\Delta\tau$ until the trial is in band, and the controller is updated only after the loop exits. It therefore only ever observes $\hat{e} \leq 1$, so $\hat{m}_e < 1$ and growth ≥ 1 by construction (Appendix B 2). Moving the band over three orders of magnitude, with gate and normalization moved together, left growth < 1 at $0/N$ at every setting. Any tolerance that serves at once as rejection threshold and control setpoint is one-sided in this way. The E1 statistics show the scale mismatch as well: the mean and median of $\hat{e}$ are 2.751×10^{-2} and 2.902×10^{-5}, far below the setpoint 1.

G3: E2 cannot resolve noise. Under the global depolarizing model the noisy flow is a deterministic rescaling of the clean flow (Eq. (10)). There is no stochastic component for an EMA to average out. The RMS ratios between Adam and Route 3 were 1.0243, 1.0246, 1.0245 and 1.0088, independent of p to ± 0.016. The E2 verdict, and a reported "stall" that came from a metric artifact, are *withdrawn.*

G6: E3 is the cap, not Adam. Table VIII gives the E3 runs. With the cap on, Route 3 and Adam are identical to every printed digit, because the cap binds on the relevant steps and neither growth law acts. The only difference appears with the cap *off.* This is a single run ($n = 1$), the report had duplicate rows, and no cost-matched column was given. The E3 advantage is therefore re-attributed to the cap and not claimed for Adam.

E1 off-by-one. The 732-circuit read-out was off by one evaluation. Corrected, the values are -1.120983 against -1.120952, and the Adam–Route 3 gap changes from $+3.5 \times 10^{-5}$ to $+3.0 \times 10^{-5}$. The correction is small: the gap between adaptive and fixed-step control at the same budget is 925× the size of this correction. At this early budget Adam is marginally ahead; read at Route 3's completion budget (Table VI) the order reverses.

G1, G7. A broken import left by a folder move was fixed (G1). Latent defects were recorded (G7), including a hard-coded SHRINK and a plot axis that clips the $s_e = 1.015$ excursion.

D. Confirmed positive result: a zero-cost error estimator (G5)

The gate already computes $\Delta E_a - \Delta E_p$. Because $Y = -\nabla E$ (Eq. (4)), $\Delta E_p = (\nabla E \cdot \dot{\theta})\Delta\tau$ is the *exact* first-order Taylor term of $E(\theta + \Delta\tau\, \dot{\theta})$. The discrepancy is therefore a second-order, local-truncation-error estimate. It costs no extra circuits, whereas an embedded pair or step doubling needs three additional f evaluations per step. The audit tested this against step doubling (one full step against two half-steps with a fresh midpoint evaluation) on the E1 trajectory:

- first 40 accepted steps: log-Pearson 0.9902, Spearman 0.9882; ratio $|\Delta E_a - \Delta E_p|/e_{\text{doubling}}$ with median 2.060 and range [0.350, 11.439];
- 20 probes: 0.9701 / median 2.157; 60 probes: 0.9948 / 2.026; all 190 steps: 0.9983 / 2.005.

The expected ratio is exactly 2. If the full-step error is $C\Delta\tau^2$, two half-steps incur $2C(\Delta\tau/2)^2 = C\Delta\tau^2/2$,

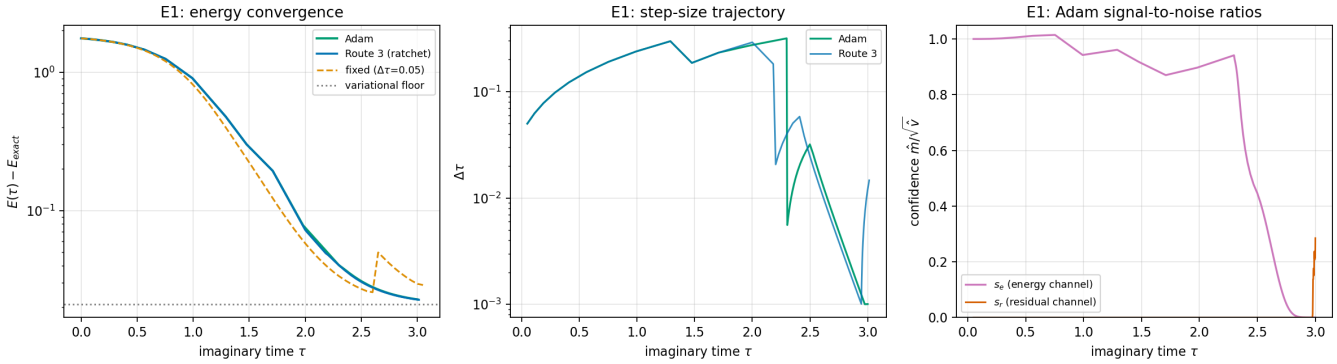


FIG. 2. E1, noiseless 5-parameter ansatz. Left: energy above the exact ground state for the Adam controller, Route 3 and fixed-step Euler ($\Delta\tau = 0.05$). The dotted line is the variational floor. Centre: step-size trajectories. Right: the Adam confidence ratios. The energy channel s_e decays to zero after $\tau \approx 2.3$, which is the "frozen" regime discussed in the text.

TABLE VI. G2: E1 at matched circuit budget on two seeds. "$E - E_{\text{floor}}$" is at the end of each run; "at budget" reads each run at the circuit count Route 3 used. $\langle s_e \rangle$ is the mean confidence ratio; "frozen" is the fraction of steps with s_e effectively zero.

Seed (budget)	Arm	Steps	Circuits	$E - E_{\text{floor}}$	at budget	$\langle s_e \rangle$	frozen
A (2069)	Route 3	172	2069	$+1.607 \times 10^{-3}$	$+1.607 \times 10^{-3}$	–	–
	Adam 0.9/0.999 (stock)	190	2285	$+1.748 \times 10^{-3}$	$+1.860 \times 10^{-3}$	0.140	49%
	Adam 0.9/0.9	179	2151	$+1.190 \times 10^{-3}$	$+1.800 \times 10^{-3}$	0.200	0%
	Adam 0.99/0.99	184	2212	$+1.437 \times 10^{-3}$	$+1.841 \times 10^{-3}$	0.355	0%
	Adam 0.999/0.999	181	2176	$+1.447 \times 10^{-3}$	$+1.803 \times 10^{-3}$	0.407	0%
D (2816)	Route 3	234	2816	$+8.555 \times 10^{-4}$	$+8.555 \times 10^{-4}$	–	–
	Adam 0.9/0.999 (stock)	500	6006	$+1.598 \times 10^{-3}$	$+4.165 \times 10^{-3}$	0.089	72%
	Adam 0.9/0.9	244	2934	$+8.641 \times 10^{-4}$	$+3.495 \times 10^{-3}$	0.394	0%
	Adam 0.99/0.99	259	3114	$+8.637 \times 10^{-4}$	$+3.942 \times 10^{-3}$	0.439	0%
	Adam 0.999/0.999	254	3054	$+8.430 \times 10^{-4}$	$+3.893 \times 10^{-3}$	0.517	0%

TABLE VII. Horizon sweep, seed D. All arms except the stock one exit at $\tau_{\max}$; the stock arm exits at the 500-step limit.

Configuration	Steps	Final E	$\langle s_e \rangle$
Route 3	234	−1.123745	–
stock 0.9/0.999	500	−1.123002	0.0893
0.999/0.999	254	−1.123757	0.5170
0.99/0.99	259	−1.123736	0.4387
0.9/0.9	244	−1.123736	0.3938
reversed 0.999/0.9	242	−1.123740	258.1220

TABLE VIII. E3: 9-parameter ansatz. Error is $E - E_{\text{exact}}$ at the end of the run.

`rcond`	Arm	Final E	Error	Steps
10^{-8}	Route 3, cap on	−1.144712	8.868×10^{-4}	108
	Route 3, cap off	−1.144201	1.398×10^{-3}	22
	Adam, cap on	−1.144712	8.868×10^{-4}	108
	Adam, cap off	−1.145034	5.653×10^{-4}	28
10^{-2}	Route 3	−1.143456	2.143×10^{-3}	17
	Adam	−1.142285	3.314×10^{-3}	14

so step doubling measures $C\Delta\tau^2/2$ while $|\Delta E_a - \Delta E_p|$ measures $C\Delta\tau^2$ (Appendix B 3). The ratio settles at 2.03–2.09 over the converged tail.

Where it fails. The ratio departs from 2 on a handful of steps (up to 11.4). The audit had predicted that these outliers would track the *change* of cond(M) along the step, since the estimator does not see the $\partial_\theta M^+$ term of $\ddot{\theta}$. This prediction *failed*. The correlation of the ratio with the level of cond(M) is −0.8938, against +0.3176 with its change, and the partial correlations are −0.5437 and +0.1523. The blind spot exists, but its predicted form does not.

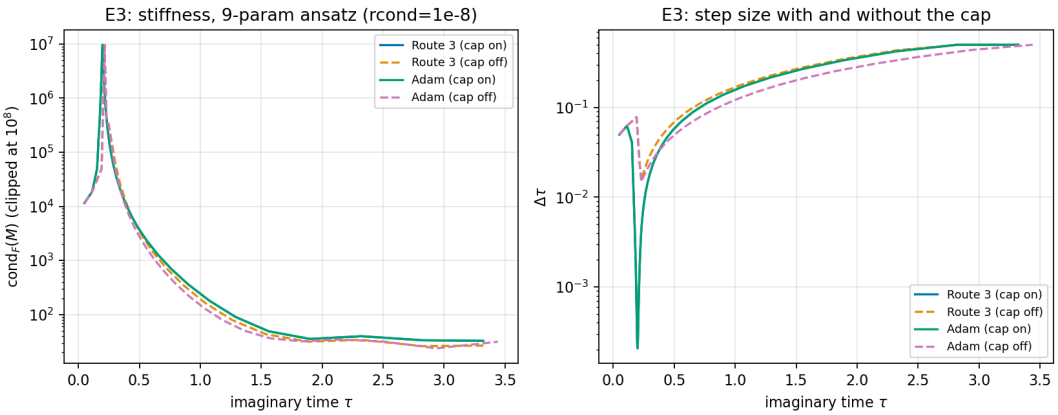


FIG. 3. E3, 9-parameter ansatz at `rcond` $= 10^{-8}$. Left: $\mathrm{cond}_F(M)$ along the four runs. It peaks near 10^7 at $\tau \approx 0.2$. Right: step size with and without the cap. With the cap on, both controllers follow the same trajectory: $\Delta\tau$ drops to about 2×10^{-4} at the peak and recovers afterwards.

V. PHASE III: ROUTE-3 CONTROLLER AND ITS AUDIT

Question. Does the memo's Route 3 (Sec. II C) work on the benchmark, and what do its indicators actually measure? The implementation defines four targets. Target A is noiseless convergence on the 5-parameter ansatz. Target B is debiasing under noise. Target C is stiffness on the 9-parameter ansatz. Target D is a noise-floor scan.

A. Audit (F1–F12)

The first write-up was audited against live code. The audit recorded twelve findings (F1–F12) and implemented six correction modules (M1–M6). The findings that changed results were:

- F1: the debiasing used the Theorem-2 factor q^{-1} instead of the hardware factor of Eq. (10).
- F3: a reported per-step number had no code path that produced it.
- F4: the RMS trajectory metric extrapolated stalled runs.
- F5: the energy gate tested against the exact-ITE model instead of the variational prediction.
- F7: Targets B and D shared one random stream, so two runs claimed to be "identical" started from different θ_0.
- F8, F9: rejected steps were committed, and comparisons were not budget-matched.
- F10: cond(M) was read through the regularizer.
- F12: a term of Eq. (3) was dropped silently. The term vanishes for pure states, and this is documented and tested in the post-audit code.

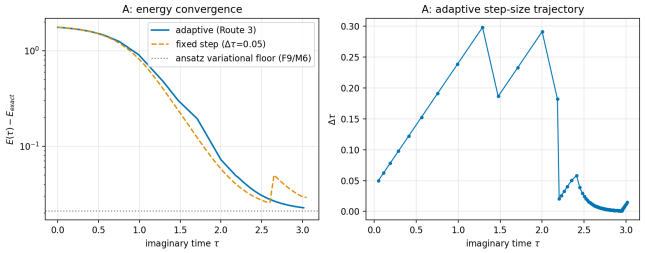


FIG. 4. Target A: Route 3 against fixed-step Euler on the noiseless 5-parameter ansatz.

The remaining findings concerned naming and code structure. F2 was the label "flow-direction error" for what is a pure magnitude rescaling, F6 a "zero violations" claim narrower than stated, and F11 a duplicated physics layer. All numbers below are post-audit.

B. Target A: convergence and cost

Route 3 reaches $E = -1.122993$ in 172 steps (Fig. 4). The fair comparison is at equal circuit count. At 732 circuits it is at -1.120952 (the corrected read-out of Sec. IV), which is 3.648×10^{-3} above E_{floor}, while fixed-step Euler is at -1.116691, 7.909×10^{-3} above. That is 2.2× closer to the floor; both gaps are differences of pinned energies. Run to completion, Route 3 ends 1.607×10^{-3} above E_{floor} after 2069 circuits, 5× closer than Euler's end point but at 2.8× its cost. Of the 2069 circuits, 1892 went to f evaluations and 177 to energy-gate trials (172 first trials and 5 retries, on 3 steps). Over the last 43 steps the cap binds on 31 and the gate retries on 0. The step collapse near $\tau_{\max}$ is therefore the stiffness cap, not the gate. Over the same steps $\sigma_{\min}$ ranges over $[3.464 \times 10^{-7}, 1.804 \times 10^{-2}]$ and the unregularized spectral cond(M) over $[32.61, 1.445 \times 10^{6}]$. The pre-audit

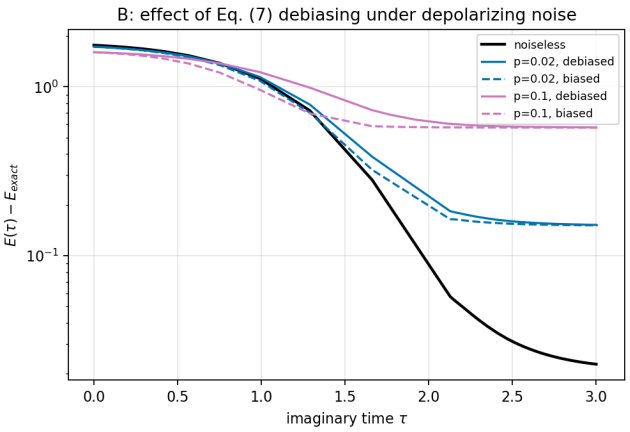


FIG. 5. Target B: energy above the exact ground state for the noiseless run and for debiased (solid) and biased (dashed) runs at $p = 0.02$ and $p = 0.1$. Both noisy pairs saturate well above the noiseless trajectory, and debiasing does not bring them systematically closer to it.

comparison (-1.1197 against -1.1167 at 61 evaluations) is superseded.

C. Target B: debiasing under noise

With the corrected factor (Eq. (10)), the debiased flow matches the clean flow to the finite-difference floor at every p (Fig. 5). The per-step error is 0.00000%, against 5.31458% (biased) at $p = 0.02$ and 34.67544% at $p = 0.1$. The noisy flow is exactly collinear with the clean one. The collinearity residuals are 1.882×10^{-8} and 1.403×10^{-8}, and the measured scale factor equals the closed form $c(p) = 1 + (1-q)/(2q)$ to six digits (1.053146 and 1.346754). Over full trajectories, however, debiasing gives no consistent advantage. The RMS deviations from the noiseless run are 1.254×10^{-1} (debiased) against 1.228×10^{-1} (biased) at $p = 0.02$, and 5.356×10^{-1} against 5.314×10^{-1} at $p = 0.1$.

The reason is structural. A scalar rescaling $\dot{\theta} \to c\,\dot{\theta}$ is indistinguishable, for any controller that is homogeneous in $\dot{\theta}$, from a faster flow, so an adaptive controller partly absorbs it as a reparametrization of τ. A single test supports this reading. The noiseless controller with every $\Delta\tau$ divided by $c(p)$ tracks the biased run better than the plain baseline does (RMS 11.9% against 12.5% at $p = 0.02$; 35.2% against 50.6% at $p = 0.1$). This test was run once and has no committed code path. The pre-audit "surprise" of a per-step debiasing error growing from about 5% to about 67% was produced by the wrong factor, and it is retracted.

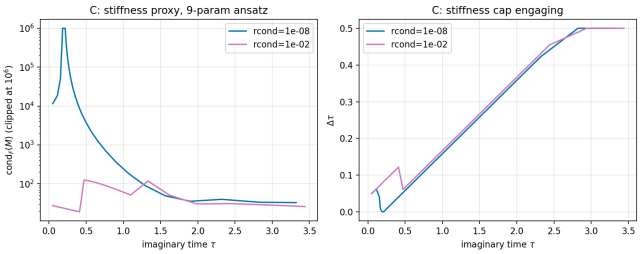


FIG. 6. Target C, 9-parameter ansatz. Left: $\mathrm{cond}_F(M)$ (clipped at 10^6) for $\texttt{rcond} = 10^{-8}$ and 10^{-2}. The loose cutoff hides the peak near $\tau \approx 0.2$. Right: step size. At the tight cutoff the stiffness cap drives $\Delta\tau$ down at the peak, and both runs then grow to $\Delta\tau_{\max}$.

TABLE IX. Target D: RMS energy deviation from the noiseless trajectory.

p	debiased	biased
10^{-4}	6.376×10^{-4}	6.067×10^{-4}
10^{-3}	6.365×10^{-3}	6.051×10^{-3}
10^{-2}	6.253×10^{-2}	5.992×10^{-2}
5×10^{-2}	2.884×10^{-1}	2.762×10^{-1}

D. Target C: the cutoff is part of the method

On the rank-6 9-parameter ansatz, the tight cutoff $\texttt{rcond} = 10^{-8}$ reaches an error of 8.9×10^{-4} in 108 steps. Along the way $\mathrm{cond}(M)$ ranges over $[32.9,\ 9.66 \times 10^6]$ and $\Delta\tau$ over $[2.07 \times 10^{-4},\ 0.5]$. The loose cutoff 10^{-2} reaches 2.1×10^{-3} in 17 steps, with $\mathrm{cond}(M)$ in $[19.1,\ 125]$ and $\Delta\tau$ in $[0.05,\ 0.5]$. The tight cutoff wins by $2.4\times$ in final error (Fig. 6). A lower reported $\mathrm{cond}(M)$ does not mean a better solve, and it is wrong to report $\mathrm{cond}(M)$ without its cutoff. Before the gate fix the same runs gave 20 steps / 1.2×10^{-2} and a 500-step stall / 2.9×10^{-2}. Most of that earlier gap came from the broken gate.

E. Target D: no noise floor

Table IX shows the RMS energy deviation from the noiseless run as a function of p. It grows smoothly and roughly linearly, with no knee. The pre-audit reading of a floor-like feature near $p \approx 10^{-3}$ rested on the stall-contaminated metric (F4) and the wrong factor (F1), and it is retracted.

VI. PHASE IV: REPLICATION OF VARIATIONAL PRODUCT FORMULAS

Why this paper. Assi *et al.* [2] evolve in *real* time with a variational product formula. The parameters (exponent coefficients) obey a McLachlan-type equation with

a metric g that plays the same role as M here. The ansatz contains repeated exponentials, so g is singular at $t = 0$ and the method needs a starting time $t_0 > 0$. This is a second, independent instance of the degeneracy faced by the controllers studied here. Table X gives the verdicts.

The figures reproduce, the text does not. Figure 4(a) reproduces to within 1.23% on all four curves, and the Ruth-to-variational ratio at $t = 1$ is 2.461. The text on the same page quotes 1.5% and 3%, which misses the figure by +41% and +73–129%. The claim that the variational formula beats Ruth's over the *entire* interval is false. The two errors scale with slopes 2.999 and 4.999 in t and cross at $t = 0.376$ (bracket 0.375–0.426). At $t = 10^{-3}$ Ruth's formula is better by up to five orders of magnitude.

Label swap and normalization. Equations (61), (65), (68) and (69) use the A and B labels swapped relative to Eq. (54). As printed, Eq. (61)→(69) misses by 2.2334 and the Eq. (65)=(68) check passes on 0 of 15 cases. With the labels swapped, the error is 3.6×10^{-16} and the check passes on 15 of 15. Figure 13 reproduces with the raw Frobenius norm but not with the paper's own Eq. (27) normalization, which differs by $2\sqrt{D} = 11.3137$.

T2′ is not scorable. The 10^{-7} floor of Fig. 2(b) is set by the integrator tolerance, which the paper does not state. The maximum error of the replication runs from 9.9×10^{-4} at `rtol` $= 10^{-3}$ down to 3.3×10^{-13} at 10^{-12}, and is 4.0×10^{-11} at the tolerance fixed before the run. A floor of 10^{-7} corresponds to an `rtol` between 10^{-8} and 10^{-6}. The number reflects the authors' integrator settings and says nothing about the method.

New result: cond(g) and its boundary layer (T6). The paper never reports cond(g), even though the method inverts g at every step. For the two-level metric of Eq. (35), cond(g) depends only on the first coefficient c_1 (Table XI). At $c_1 = 0$ the matrix drops to rank 2 of 3 with null vector $\propto (1, 0, -1)$, because the two A exponentials of the ABA ansatz become indistinguishable. For $c_1 \in [10^{-6}, 10^{-3}]$ the fitted exponent is cond(g) $\propto c_1^{-2.0000}$. Along the QIM trajectory cond(g) starts at 3.4×10^{12} (BAB) and 2.1×10^{16} (ABAB) and falls to 1.36 and 4.85 at $t = 2$ (Fig. 7). The ill-conditioning is real but confined to a boundary layer, and the result is insensitive to how it is regularized. Varying $t_0 \in [10^{-8}, 10^{-3}]$ and `rcond` $\in [10^{-14}, 10^{-4}]$ moves $\mathcal{E}_F(t{=}2)$ by at most 5.12×10^{-10} relative.

VII. PHASE V: PRE-REGISTERED EXPERIMENTS P1 AND P2

Phase V consists of two experiments, each with written predictions registered before it was run. P1 asks what the cond(M) $\sim 10^7$ events of Target C and E3 are. P2 asks whether a step law with an explicit tolerance controls the achieved error.

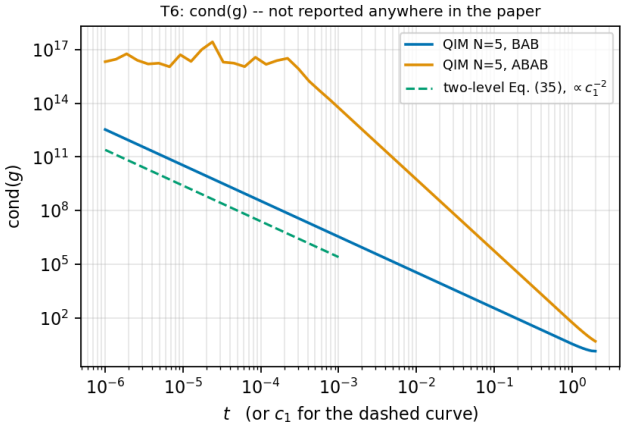


FIG. 7. T6: cond(g) along the QIM ($N = 5$) trajectory for the BAB and ABAB ansätze, and the two-level closed form (dashed, against c_1), which scales as c_1^{-2}.

A. P1: classifying the singularity

Hypotheses. The large cond(M) could come from (a) gauge leakage, meaning the three structural null directions of the rank-6 metric passing the cutoff, (b) finite-difference noise in M, or (c) a genuine singularity of the parametrization.

Predictions and outcome. Table XII scores the predictions.

No gauge leakage. The largest σ_7/σ_1 is 2.021×10^{-16} at `rcond` $= 10^{-8}$ and 1.321×10^{-16} at 10^{-2}. At 10^{-8} the pseudo-inverse always keeps exactly 6 modes. At 10^{-2} it keeps 4 to 6, and fewer than 6 on 11 of 17 steps. The rank-6 condition number equals the logged value to 1.11×10^{-15}. The gauge singular values sit lower than predicted because M is a Gram matrix. A true null vector picks up a contribution that is *second* order in the finite-difference error, about 10^{-20}, so what remains is the eigensolver's roundoff at about 10^{-16}. The per-entry error $\max|M_{\mathrm{fd}} - M_{\mathrm{exact}}|$ is 1.652×10^{-10} (1.715×10^{-10} at the loose cutoff).

Identities. The state form Eq. (5) agrees with the finite differences to 2.594×10^{-10} for M and 3.955×10^{-10} for Y. The opposite sign convention gives 1.589, so the sign is fixed unambiguously. The CRZ row of M is exactly zero at $\theta = 0$, and Eq. B9 of Ref. [1] holds to 7.216×10^{-16}.

A chart singularity. The peak is at step 43, $\tau = 0.1983$, with singular values

$$[0.9991,\ 0.9828,\ 0.1463,\ 0.09246,\ 0.02142,\ 1.461\times10^{-7},\ 1.4\times10^{-16},\ 3.3\times10^{-17},\ 8.8\times10^{-19}].$$

The soft mode v_6 loads on a single gate (th8, weight +0.934; th3 +0.233, th2 −0.172). At this point $\langle Z_0\rangle = 0.9944$ and $M_{88} = 5.568\times10^{-3}$. The gates pair up in the *gauge* vectors instead (Fig. 9): $v_7 =$ th1−th6 (∓0.693), $v_8 \approx$ th2−th7 ($+0.638/-0.609$), $v_9 \approx$ th4 (-0.998).

TABLE X. Phase IV verdicts for Ref. [2]. "QIM" is the quantum Ising model with $N = 5$ used in the paper. Read-off precision from figures is ± 0.001.

Target	Paper	This work	Verdict
T1 gate counts, $N = 2$–12	$R_x = 2N$, $R_z = 4N - 2$, CNOT $4N - 4$	identical	*reproduced*
T1 circuit vs `expm`, computed angles / Fig. 12 angles	exact	2.8×10^{-16} / 1.7286	*reproduced* / *not*
T2 two-level model, Eqs. (31)–(36)	exact	$\leq 2.8 \times 10^{-14}$	*reproduced*
T2′ Fig. 2(b) error floor	"of order 10^{-7}"	4.0×10^{-11}	inconclusive
T3 $\mathcal{E}_F(t{=}1)$, four curves of Fig. 4(a)	0.0212, 0.0212, 0.0692, 0.0514	0.021217, 0.021143, 0.068847, 0.052034	*reproduced* ($\leq 1.23\%$)
T3′ same, as stated in the text	1.5% / 3%	0.0211 / 0.0688	*not reproduced*
T3″ variational beats Ruth on all of $[0, 2]$	true	false	*not reproduced*
T4 Eq. (61)→(69), labels as printed / $A \leftrightarrow B$	exact	2.2334 / 3.6×10^{-16}	*not* / *reproduced*
T4 Fig. 13 at $t = 100$, raw norm	0.2694, 0.1715	0.26951, 0.171821	*reproduced*
T4 same, normalized by Eq. (27)	0.2694, 0.1715	0.023822, 0.015187	*not reproduced*
T5 Eq. (65)=(68), printed / $A \leftrightarrow B$	15/15	0/15 / 15/15	*not* / *reproduced*
T5 Ξ predicts the better ordering	7/7	0/7 literal, 7/7 relabelled	only relabelled
T7 stroboscopic $\mathcal{E}_F(t{=}100)$	0.11, 0.055	0.108968, 0.055305	*reproduced*
T8 advantage grows with τ	true	1 of 3 statistics	*partially reproduced*

TABLE XI. T6: conditioning of the two-level metric g of Ref. [2], Eq. (35).

c_1	$\det g$	rank	cond(g)
0	0	2 of 3	2.2×10^{17}
10^{-6}	4.0×10^{-8}	3	2.50×10^{11}
10^{-3}	4.0×10^{-2}	3	2.50×10^{5}
0.2	1.2865×10^{3}	3	10.60

TABLE XII. P1 scorecard.

Registered prediction	Outcome
$\sigma_{7\text{–}9}$ below the $10^{-8}\sigma_1$ cutoff (no leakage)	held
$\sigma_{7\text{–}9}$ at 10^{-10}–10^{-12}	failed (favourably)
rank-6 cond equals logged cond along E3	held
the cap reads real stiffness	held
the soft mode is a near-degenerate gate *pair*	failed

TABLE XIII. P1: speeds along the E3 trajectory (`rcond` = 10^{-8}). The last column is the fraction of $\dot\theta$ along the soft singular vector v_6.

τ	cond	$\|\dot\theta\|$	$\|\dot\rho\|$	along v_6
0	1.156×10^{4}	0.7989	0.1428	0.351
0.1880	2.492×10^{6}	35.23	0.1826	0.99979
0.1949	7.392×10^{6}	98.8	0.1848	0.999992
0.1983	9.664×10^{6}	128.4	0.1862	0.9999978
0.2016	7.476×10^{6}	100.2	0.1877	0.999989
2.8223	32.85	0.6401	0.257	0.276

Table XIII and Fig. 8 are the decisive evidence. Near the peak the parameter speed $\|\dot\theta\|$ grows by two orders of magnitude, and almost all of it lies along v_6, while the *state* speed $\|\dot\rho\|$ barely moves. Over the whole run $\|\dot\theta\|$ varies by a factor of 160.8 and $\|\dot\rho\|$ by only 8.83. The state moves smoothly through a region where the chart $\theta \mapsto \rho$ folds, so a small state displacement needs a large parameter displacement. This is a coordinate singularity, and it is real stiffness in the sense relevant to an explicit integrator. The cond(M) cap therefore reads a genuine property, and the attribution of E3 to the cap (G6) is supported. There are 86 of 108 steps with cond $> 10^5$ at `rcond` $= 10^{-8}$, against 0 of 17 at 10^{-2}. The largest logged cond is 9.6637×10^6 against 125.39, and the largest spectral value 6.8367×10^6 against 1.9281×10^5.

The fold amplifies finite-difference error, but only locally. Re-running with the exact state-form metric changes almost nothing. At `rcond` $= 10^{-8}$ the final error is 8.868181×10^{-4} (finite differences) against 8.854194×10^{-4} (exact), both in 108 steps, with largest cond 9.66375×10^6 against 9.66367×10^6. At 10^{-2} both runs give 2.143405×10^{-3}. The energy gap between the two runs is 1.292×10^{-11} before the fold. It peaks at 1.101×10^{-4} ($\tau = 2.3223$) and ends at 6.068×10^{-6}. At the loose cutoff the gap stays at or below 1.228×10^{-9}.

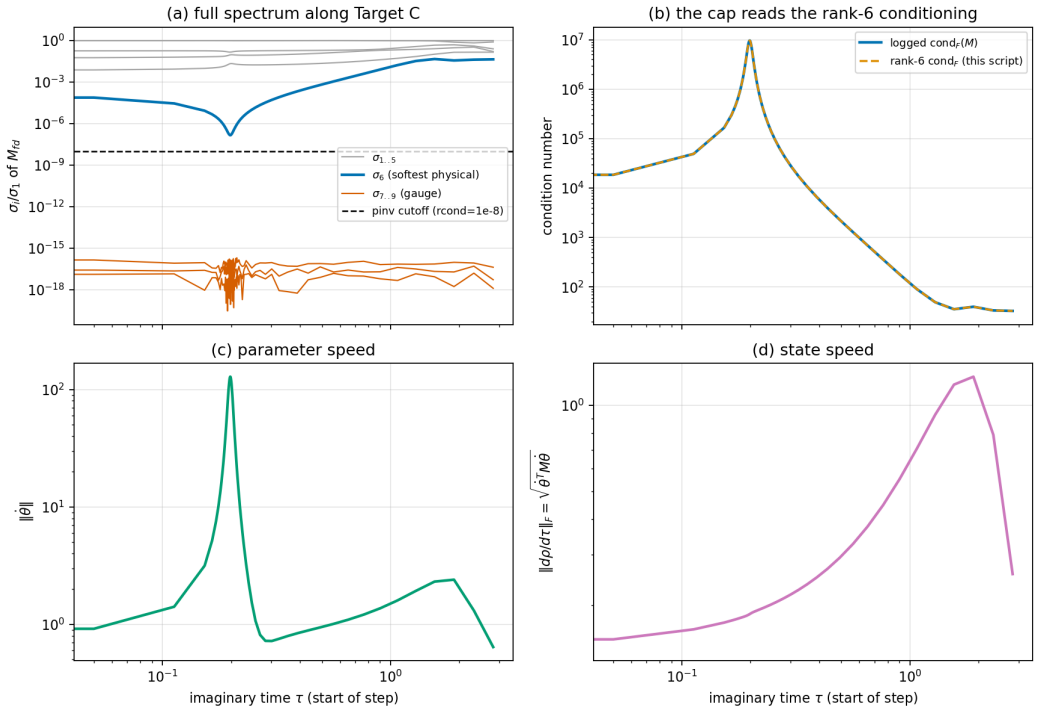


FIG. 8. P1, Target C/E3 run at `rcond` $= 10^{-8}$. (a) Singular values of M relative to σ_1. The three gauge modes stay at roundoff, far below the cutoff (dashed). The soft physical mode σ_6 dips to about 10^{-7} at $\tau \approx 0.2$. (b) The logged cond$_F(M)$ and the rank-6 value coincide. (c) The parameter speed spikes at the peak. (d) The state speed $\|\dot\rho\|_F = \sqrt{\dot\theta^\top M \dot\theta}$ does not.

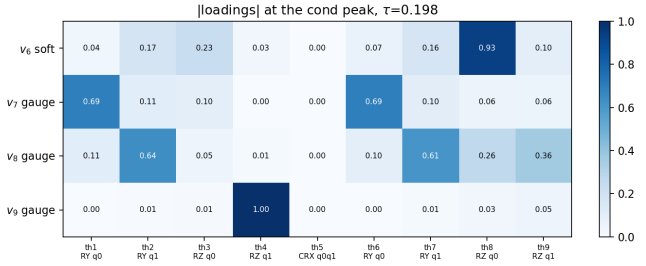


FIG. 9. P1: absolute loadings of the soft vector v_6 and the gauge vectors $v_{7\text{–}9}$ on the nine gates at the cond peak, $\tau = 0.198$.

A second mechanism on the 5-parameter ansatz. As a control, Target A (error 2.2606×10^{-2} in 172 steps) keeps only 4 modes on 12 of 172 steps, all in $\tau \in [2.945, 2.999]$. There the soft vector is the CRZ gate alone (+0.999) with $\sigma_5/\sigma_1 = 6.921\times10^{-7}$ at $\tau = 2.9987$. The state is almost a computational basis state, with populations 7.53×10^{-4}, 1.74×10^{-7}, 0.99902 and 2.30×10^{-4} and $\langle Z_1\rangle = 0.99954$. A controlled rotation whose control qubit is almost deterministic has almost zero generator variance, so its metric entry vanishes ($M_{\mathrm{CRZ}} = 8.3662\times10^{-4}$). This is a zero-variance mechanism, distinct from the fold. $\|\dot\theta\|$ varies by 116.7× here.

TABLE XIV. P1: cap fed by the logged or the rank-6 cond.

`rcond`	cap reads	error	steps	binding
10^{-8}	logged	8.8682×10^{-4}	108	82
10^{-8}	rank-6	8.8617×10^{-4}	108	82
10^{-2}	logged	2.1434×10^{-3}	17	0
10^{-2}	rank-6	3.4051×10^{-3}	105	67

A regime-aware cap is rejected. Feeding the cap the rank-6 condition number instead of the logged one was tested (Table XIV). At the tight cutoff it changes nothing. At the loose cutoff it makes the cap bind on 67 steps where it bound on none, and the error rises from 2.1434×10^{-3} to 3.4051×10^{-3}. The variant was rejected.

B. P2: does the achieved error track a requested tolerance?

Set-up. Neither controller of Phases II–III has a tolerance to sweep. Route 3 ratchets, and Adam's setpoint is the reject band (G4). P2 therefore drives the step with the PI law Eq. (8) ($k_I = 0.15$, $k_P = 0.2$), fed by the zero-cost estimator of Sec. IV D, $e_n = |\Delta E_a - \Delta E_p|/2$. The reject band stays as the gate. Two error-per-step policies are compared: EPS ($e_n \le$ tol) and EPUS

($e_n/\Delta\tau \le$ tol). For an order-1 method they predict end-point error $\propto$ tol$^{1/2}$ and $\propto$ tol^1 respectively (Appendix B 4). The reference is an exact-metric DOP853 solution at `rtol` $= 10^{-11}$, with self-check 8.25×10^{-12}. A run qualified for the fit only if all of the following held, as registered in advance: it exits at $\tau_{\max}$, the cap binds on fewer than 25% of steps, $\Delta\tau$ sits at a bound on fewer than 25% of steps, and the error is more than 10× the reference floor. At least three qualifying runs were required.

Outcome. Only 2 of 9 EPS runs and 2 of 8 EPUS runs qualified, so no exponent could be fitted. Five of the six registered predictions (the three exponent windows, the location of the break, and the circuit slope) therefore score as *failed for lack of a fit*. None of them failed because a slope was measured outside its window. The prediction that did hold is the one that tests G4: with a tolerance separate from the band, growth < 1 occurs in every run (527 of 1234 steps at EPS tol $= 10^{-6}$). Where tol lies below the estimator's natural scale, the median of e_n/tol is 0.99. The PI law does control e_n. It is the end-point error that fails to follow.

Table XV and Fig. 10 give the full sweeps. The local slopes of log err against log tol between neighbouring runs wander from −0.80 to +2.62: +0.47, +0.25, −0.07, −0.42, −0.80, +1.61, +2.13, −0.45 for EPS and +0.04, −0.20, −0.64, +2.62, +0.59, +0.34, +1.08 for EPUS. The cap takes over at tol $= 10^{-5}$ (EPS) and 3.16×10^{-3} (EPUS), with first binding between $\tau = 2.091$ and 2.409. For reference, under the same metric Route 3 ends at 6.3181×10^{-4} (172 steps, 2069 circuits) and Adam at 7.2773×10^{-4} (190 steps, 2285 circuits). Replaying Adam inside the P2 harness reproduces its E1 run bit for bit (−1.122851752157184, 190 steps, 2285 circuits).

Why the end-point error does not track tol *(post hoc).* This analysis was not registered in advance. Two mechanisms act together. First, every run starts with the fixed $\Delta\tau_0 = 0.05$, whose own error is $e_1 = 2.2894 \times 10^{-4}$, which is 229 × tol at tol $= 10^{-6}$. This first step contributes 5.2914×10^{-4} to the error whatever tol is. Second, the flow is contractive. The error along each run peaks at $\tau \approx 0.94$–1.15, and the dynamics then damp it by two to three decades before τ_{end}. The error at the end is therefore the damped remnant of early, tol-independent errors, not an accumulation of per-step errors controlled by tol. All runs end between 8.3×10^{-4} and 3.45×10^{-3} above E_{floor}. The reference itself sits 1.0212×10^{-3} above it at $\tau = 3$.

First-step ablation (post hoc). Starting every run at the minimum step instead of $\Delta\tau_0$ restores the expected scaling of the maximum error over $\tau \le 2$, the part of each run before the cap engages (Table XVI, Fig. 11). The largest error along the run drops from 6.977×10^{-3} to 2.469×10^{-3} (EPS, tol $= 10^{-6}$) and from 8.413×10^{-3} to 2.057×10^{-3} (EPUS, 3.16×10^{-4}). The end-point error stays between 2.0×10^{-5} and 5.1×10^{-5} because the contraction still dominates it.

The local slopes with the ablated first step decline smoothly toward the cap: 0.48, 0.47, 0.46, 0.43, 0.40, 0.34, 0.26, 0.19 (EPS) and 0.94, 0.84, 0.71, 0.49, 0.24, 0.05, 0.00 (EPUS).

VIII. SYNTHESIS

A. Status of the claims

Table XVII lists every substantive claim the project has made, with its present status and where the evidence sits.

B. Four ways the metric degenerates

The project observed four distinct forms of degeneracy of the McLachlan metric. They call for different treatment, and a single number such as cond(M) does not tell them apart.

1. **Structural rank deficiency.** The ansatz has more parameters than the manifold has dimensions. The 9-parameter ansatz has exactly three gauge modes at roundoff throughout (T3, P1). They are harmless if the cutoff removes them, and they make any unregularized cond(M) meaningless.
2. **Boundary layer.** Repeated generators coincide at a special point, $t = 0$ in Ref. [2], with cond(g) $\propto c_1^{-2}$. The layer is confined, and the result does not depend on how it is regularized (T6).
3. **Chart singularity (fold).** The map $\theta \mapsto \rho$ folds along the trajectory. One soft mode reaches $\sigma \sim 10^{-7}$, the parameter speed diverges and the state speed stays bounded (P1). This is the stiffness the Route-3 cap handles.
4. **Zero-variance generator.** A gate whose generator has vanishing variance in the current state, here a CRZ with an almost deterministic control, drops out of M (P1, Target A control).

C. How the problems were found and handled

The project's corrections share a pattern. Every retraction in Table XVII came from a number that had been produced once and never re-derived by code: the debias factor, the per-step "surprise", the noise-floor reading and the s_e bound. Each was caught by an audit that re-ran the claim against live code. The response has been procedural. Each claim has one function, every quoted value is pinned with no adjustable tolerance, and from Phase V on predictions are registered before the run. Failed predictions are reported as failures (P1: two of five; P2: five of six, for lack of a fit). Post-hoc analyses are labelled as such.

TABLE XV. P2 tolerance sweeps. "err" is $|E(\tau_{\rm end}) - E_{\rm ref}|$; "max err" is the maximum along the run; "cap" is the fraction of steps on which the stiffness cap binds; "shrink" counts steps with growth < 1; the last column is the median of e_n/tol. Qualifying runs are the first two of each arm.

Arm	tol	err($\tau_{\rm end}$)	max err	steps	circuits	cap %	shrink / med. e_n/tol
EPS	10^{-6}	1.1067×10^{-5}	6.977×10^{-3}	1234	14808	10.8	527 / 0.992
	3.16×10^{-6}	1.9022×10^{-5}	9.066×10^{-3}	761	9132	18.4	295 / 0.974
	10^{-5}	2.5242×10^{-5}	1.291×10^{-2}	497	5964	29.0	163 / 0.897
	3.16×10^{-5}	2.3316×10^{-5}	1.917×10^{-2}	351	4212	41.6	91 / 0.775
	10^{-4}	1.4299×10^{-5}	2.918×10^{-2}	270	3240	54.4	52 / 3.35×10^{-3}
	3.16×10^{-4}	5.6702×10^{-6}	4.468×10^{-2}	225	2700	65.3	29 / 2.61×10^{-4}
	10^{-3}	3.6387×10^{-5}	6.479×10^{-2}	200	2400	73.5	14 / 4.75×10^{-5}
	3.16×10^{-3}	4.2490×10^{-4}	8.954×10^{-2}	184	2208	79.9	6 / 9.5×10^{-6}
	10^{-2}	2.5438×10^{-4}	1.130×10^{-1}	180	2162	80.0	2 / 2.9×10^{-6}
EPUS	3.16×10^{-4}	2.9473×10^{-5}	8.413×10^{-3}	2260	27120	6.2	1110 / 1
	10^{-3}	3.0816×10^{-5}	1.420×10^{-2}	822	9864	17.8	339 / 0.985
	3.16×10^{-3}	2.4532×10^{-5}	2.751×10^{-2}	371	4452	39.6	105 / 0.633
	10^{-2}	1.1781×10^{-5}	5.366×10^{-2}	228	2736	64.9	30 / 2.43×10^{-3}
	3.16×10^{-2}	2.4098×10^{-4}	8.966×10^{-2}	186	2233	78.0	7 / 4.07×10^{-4}
	0.1	4.7659×10^{-4}	1.209×10^{-1}	176	2115	81.8	2 / 1.08×10^{-4}
	0.316	7.0607×10^{-4}	1.325×10^{-1}	172	2068	83.1	2 / 3.23×10^{-5}
	1	2.4412×10^{-3}	1.325×10^{-1}	56	676	67.9	1 / 1.67×10^{-4}

TABLE XVI. P2 first-step ablation (post hoc): slopes on the pre-cap part of the sweep against the order-1 prediction.

Arm	slope, $\Delta\tau_0$	slope, min. step	theory	circuit slope
EPS	0.267	0.478	0.5	-0.385 [-0.5]
EPUS	0.515	0.890	1	-0.765 [-1]

IX. SCOPE OF THE EVIDENCE

The results above hold under the following conditions, which are stated so that a reader can calibrate them.

- **One system.** All controller results are for H_2/STO-3G on two qubits with the two QDP ansätze. The Phase IV results are for the models of Ref. [2].
- **No shot noise.** All expectation values are exact. The depolarizing model is deterministic, which is why E2 was withdrawn.
- **Finite-difference derivatives** ($h = 10^{-6}$), validated to about 10^{-10} and, in P1, against the exact state form. The finite-difference error reaches 1.1×10^{-4} in energy after the fold.
- **Chosen constants.** Every threshold in Table II was chosen for this work, and the conclusions are specific to them.
- **Few seeds.** G2 uses two seeds, and E3 is a single run.
- **Unresolved discrepancies.** The QDP condition numbers (T3) and the Fig. 3(b) reading (T5) have no identified cause on the paper's side. An error on the replication side is excluded as far as the 60-row convention scan allows.
- **Unpinned checks.** The reparametrization test of Target B and a few side checks in Phase IV were run once and are labelled as such in the source notes.

Appendix A: Reproducibility

All code runs on Python 3.11 with PennyLane 0.45.1, NumPy and SciPy, in the project's `qml-julia` environment. Paths are relative to the project folder.

A `-check` run recomputes every pinned value and fails on any change. There is no tolerance argument, so a deliberate perturbation of a pinned value makes the check fail.

Appendix B: Derivations

1. The factor of two

Let $\rho' = q\rho + (1-q)\mathbb{1}/d$ denote both the differentiated state and the state inside $\{H, \rho\}$. Then $\partial_k \rho' = q\,\partial_k \rho$ and $M' = q^2 M$. For Y,

$$\{H, \rho'\} - 2\langle H\rangle' \rho' = q\{H, \rho\} + \tfrac{2(1-q)}{d} H - 2\langle H\rangle' \rho'.$$

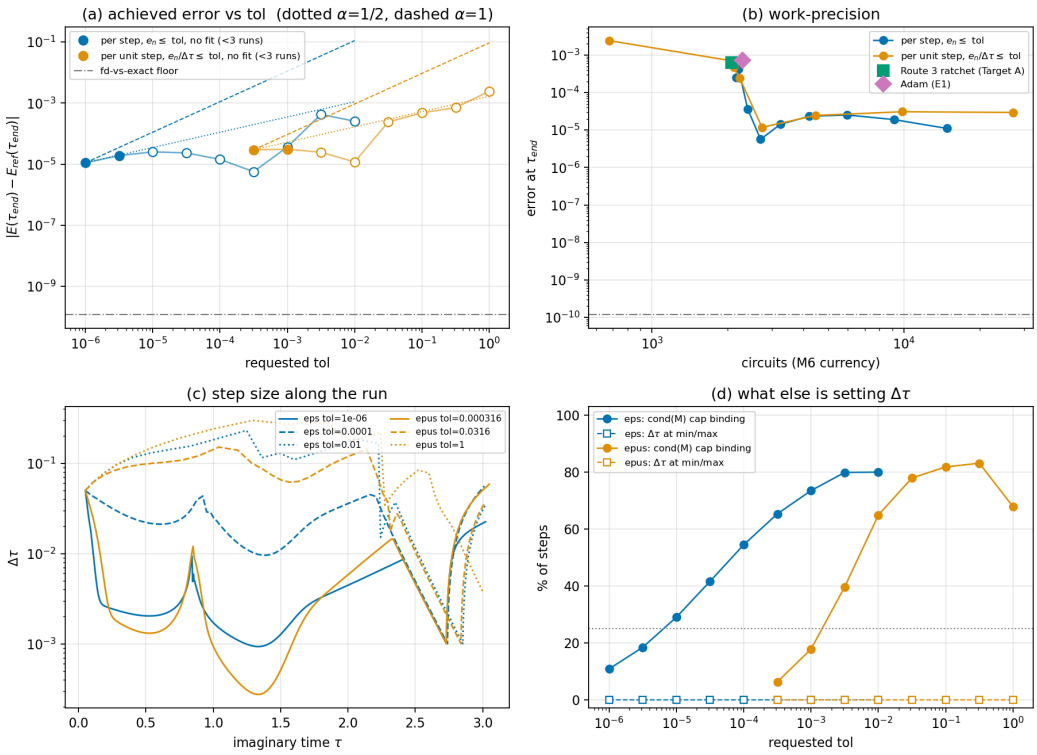


FIG. 10. P2 as registered. (a) End-point error against requested tolerance. Filled markers are runs that qualify for the fit; the dotted and dashed lines are the $\alpha = 1/2$ and $\alpha = 1$ predictions. (b) Work–precision in circuits, with the Route 3 and Adam anchors. (c) Step size along selected runs. (d) Fraction of steps on which the cond(M) cap binds or $\Delta\tau$ sits at a bound. The dotted line is the 25% qualification limit.

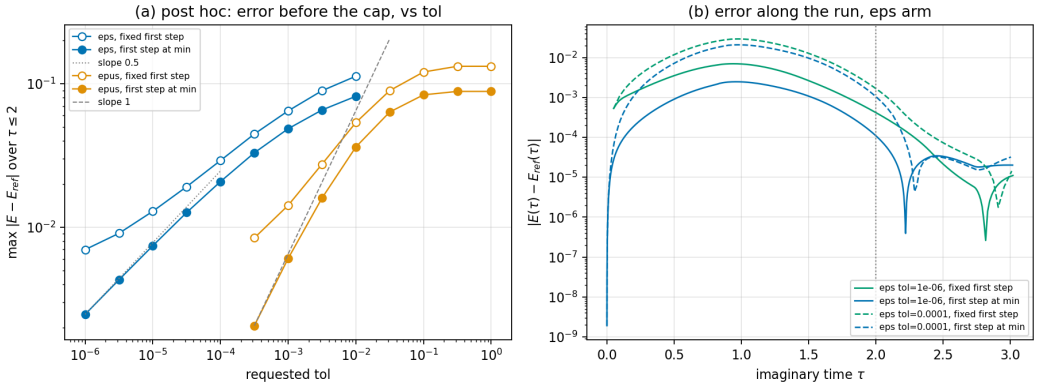


FIG. 11. P2 first-step ablation (post hoc). (a) Maximum error over $\tau \le 2$, before the cap engages, against tol, with the fixed first step (open) and with the first step at the minimum (filled). (b) Error along two EPS runs. The error peaks near $\tau \approx 1$ and is then damped by the flow.

Contracting with $\partial_k\rho' = q\,\partial_k\rho$ removes the last term because $\mathrm{Tr}[\partial_k\rho\,\rho'] = q\,\mathrm{Tr}[\partial_k\rho\,\rho] + \frac{1-q}{d}\mathrm{Tr}\,\partial_k\rho = 0$ (for a pure state $\mathrm{Tr}[\partial_k\rho\,\rho] = \frac{1}{2}\partial_k\mathrm{Tr}\rho^2 = 0$). Since $\mathrm{Tr}[\partial_k\rho\{H,\rho\}] = \partial_k E = -Y_k$ and $\mathrm{Tr}[\partial_k\rho\,H] = \partial_k E$,

$$Y_k' = q^2 Y_k + \frac{2q(1-q)}{d} Y_k.$$

Hence $\dot\theta' = M'^{+}Y' = \dot\theta\,[1 + 2(1-q)/(dq)]$. For $d = 4$ this is $\dot\theta[1 + (1-q)/(2q)]$. If only $\partial_k\rho$ is noisy, as in Ref. [1], $Y' = qY$ and $\dot\theta' = \dot\theta/q = \dot\theta[1 + (1-q)/q]$, which is Theorem 2.

2. The update law cannot shrink the step

The gate rejects exactly when $\hat e > 1$ (Eq. (11)), and the controller is updated only on accepted trials. Every

TABLE XVII. Status of the project's claims at the close of Phase V.

Claim	Status	Evidence
Theorem 1 and the noisy-flow scaling of Ref. [1] hold	established	Sec. III, T1/T4
Theorem 2 overstates the hardware velocity error by exactly 2×	established	Eq. (10), Fig. 1
QDP condition numbers (66.72; 20.22)	not reproduced	Sec. III, T3
9-parameter metric is structurally rank 6 of 9	established	T3, P1
Route 3 is 2.2× closer to the floor than fixed Euler at 732 circuits	established	Target A
Debiasing with the corrected factor is exact per step	established	Target B
Debiasing improves full trajectories	not supported	Target B, Target D
Per-step debiasing error of 5–67% ("surprise")	retracted	F1/M1
Noise floor near $p \approx 10^{-3}$	retracted	F4, Target D
Tight `rcond` beats loose despite higher cond(M)	established	Target C, P1
Adam-style memory improves the step law	not supported	G2, Table VI
Adam is robust to noise (E2)	withdrawn	G3
Adam improves stiffness handling (E3)	re-attributed to the cap	G6
$s_e \leq 1$	false	G2
Energy-gate discrepancy is a zero-cost LTE estimator (ratio 2)	established	G5
Estimator blind spot follows Δcond(M)	prediction failed	G5
Route-3 stiffness is a chart singularity, not leakage or noise	established	P1
PI law with explicit tol controls e_n	established	P2
End-point error scales as tol^{α}	not fittable	P2
Fixed first step and contraction explain the failure	post hoc	P2 ablation
Figures of Ref. [2] reproduce; four internal inconsistencies found	established	Table X

TABLE XVIII. Scripts and pinned values.

Phase	Script	Pinned / check
I	`Replications/practical-limitations-qdp/replicate.py`	one function per target
II	`Adam_QITE/adam_qite.py`, `verify_audit.py`	147 values, 0 drift, ≈28 s
III	`Attack_Route_03/adaptive_qite.py` (M1–M6)	`check_eq_b9_self`, per-target seeds
IV	`Replications/Beyond_Trot/replicate.py -check`	2 runs byte-identical
V	`Route_03_and_Beyond_Trot/Proposal_1/p1_singularity.py`	51 values, 0 drift
V	`Route_03_and_Beyond_Trot/Proposal_2/p2_tol_proportionality.py`	93 values, 0 drift, byte-identical reruns

observed $\hat{e}$ is therefore at most 1, and so are the EMAs $\hat{m}_e$. With $s_e, k_I \geq 0$, $\max(\hat{m}_e, \varepsilon)^{-s_e k_I} \geq 1$. The growth factor is at least 1 on every step, independently of the band.

3. Ratio two against step doubling

Let one Euler step of size $\Delta\tau$ have local error $C\Delta\tau^2$. Two steps of $\Delta\tau/2$ have $2C(\Delta\tau/2)^2 = C\Delta\tau^2/2$, so the difference between the two, which is what step doubling reports, is $C\Delta\tau^2/2$. The gate's $\Delta E_a - \Delta E_p$ is the full second-order remainder of the energy along the Euler step, $C\Delta\tau^2$ read in energy. The ratio is therefore 2. The remainder contains $\frac{1}{2}\Delta\tau^2\,\dot{\theta}^\top \nabla^2 E\,\dot{\theta}$ but not the part of $\ddot{\theta}$ that comes from $\partial_\theta M^+$, which is the origin of the blind spot in Sec. IV D.

4. Tolerance exponents for EPS and EPUS

For a method of order p, the local error is $e_n \approx C\Delta\tau^{p+1}$. Under EPS the controller holds $e_n \approx \text{tol}$, so $\Delta\tau \propto \text{tol}^{1/(p+1)}$. The global error, which is about $T\,C\Delta\tau^p$, then scales as $\text{tol}^{p/(p+1)}$, and the step count as $\text{tol}^{-1/(p+1)}$. Under EPUS it holds $e_n/\Delta\tau \approx \text{tol}$, so $\Delta\tau^p \propto \text{tol}$ and the global error scales as tol^1, with step

count $\propto \mathrm{tol}^{-1/p}$. For explicit Euler ($p = 1$) the predictions are $\mathrm{tol}^{1/2}$ with circuit slope $-1/2$ (EPS) and tol^{1} with circuit slope -1 (EPUS). These are the "theory" values in Table XVI.

Appendix C: Anticipated questions

Why finite differences and not parameter-shift rules? The derivative $\partial_k \rho$ of a density matrix is needed for M and Y, and finite differences give it for any gate, including under noise. The results were validated against the closed-form diagonal to 8.9×10^{-11} and, in P1, against the exact state form. The only place the difference matters is after the fold, where it reaches 1.1×10^{-4} in energy along the run; the final error changes from 8.868181×10^{-4} to 8.854194×10^{-4}.

Why quote cond_F *at all, if it depends on the cutoff?* Because the controller acts on the regularized system that it actually solves. P1 shows that, at the tight cutoff, the logged value equals the rank-6 condition number to 10^{-15}, so it reads the physical soft mode. The report always gives the cutoff with the number.

Is the stiffness physical or an artifact of the ansatz? It is a property of the parametrization, a fold of $\theta \mapsto \rho$, and not of the state trajectory, whose speed stays bounded (Table XIII). For an explicit integrator in θ it is real stiffness all the same, and it caps the stable step.

Could the replication gaps be implementation bugs? The same code reproduces the closed-form targets of both papers to 10^{-10}–10^{-16}. Each gap is one-sided or structural (T2, T3), survives a scan over conventions (60 rows for T3), or is resolved by a single identifiable change on the paper's side (the $A \leftrightarrow B$ swap, the Eq. (27) normalization).

Why does Adam act only on $\Delta\tau$*?* Rescaling parameters individually changes the direction of the update, so the integrator would no longer follow McLachlan's flow. The controller was restricted to the one quantity a step controller is allowed to change.

Why no shot noise? The questions studied here (conditioning, step law, bias factor) are deterministic, and adding shot noise would hide them. As a consequence, no claim about robustness to stochastic noise is made (G3).

Why does the end-point error fail as an accuracy measure? Imaginary-time flow is contractive. Errors made early are damped by two to three decades before the end (Sec. VII), so the final error measures the damped remnant of early errors and not the local accuracy the controller maintains.

Why is Route 3 still the baseline, if it has no tolerance? It wins every matched-budget comparison against Adam (Table VI), and its constants are fixed and documented. Several PI runs of P2 end closer to the reference than Route 3 at similar cost (Table XV; e.g. 4.7659×10^{-4} at 2115 circuits against 6.3181×10^{-4} at 2069), but on one seed and with an end-point error that is not monotone in the tolerance, so no advantage is claimed. A second seed at matched circuits (P2b) is the test.

[1] G. Saxena, A. Shalabi, and T. H. Kyaw, Practical limitations of quantum data propagation on noisy quantum processors, Phys. Rev. Applied **21**, 054014 (2024), arXiv:2306.13046.

[2] I. Assi, M. Vogl, M. Kumari, and J. P. F. LeBlanc, Beyond Trotterization: Variational product formulas for quantum simulation, Phys. Rev. B **113**, 214314 (2026), arXiv:2511.15124.

[3] A. Anglés-Castillo, L. Ion, T. Pandit, R. Gomez-Lurbe, R. Martínez, and M. A. Garcia-March, Understanding quantum imaginary time evolution and its variational form, arXiv:2510.02015 (2025).

[4] A. Arulandu, J. M. Martyn, and I. L. Chuang, Trading imaginary time for randomness in ground state preparation, arXiv:2608.00443 (2026).

[5] M. Motta, C. Sun, A. T. K. Tan, M. J. O'Rourke, E. Ye, A. J. Minnich, F. G. S. L. Brandão, and G. K.-L. Chan, Determining eigenstates and thermal states on a quantum computer using quantum imaginary time evolution, Nat. Phys. **16**, 205 (2020).

[6] S. McArdle, T. Jones, S. Endo, Y. Li, S. C. Benjamin, and X. Yuan, Variational ansatz-based quantum simulation of imaginary time evolution, npj Quantum Inf. **5**, 75 (2019).

[7] Adaptive time-stepping for variational QITE: An RKF-type controller with a hardware-noise floor, internal memo, LG Electronics Toronto AI team (unpublished).